

Daniel Daza

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🎓 [Google Scholar](#)

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Summary	My research focuses on artificial intelligence methods for representing and learning from structured scientific data such as graphs. This includes methods for constructing structured representations , such as building and maintaining knowledge graphs from heterogeneous and textual data using language models, and methods for learning from structured representations , including foundation models that combine knowledge graphs with multimodal scientific data to support prediction, reasoning, and scientific discovery.	
Education	Vrije Universiteit Amsterdam	September 2019 - June 2024
	<i>Doctor of Philosophy, cum laude</i>	Amsterdam, The Netherlands
	Thesis: Exploiting Subgraphs and Attributes for Representation Learning on Knowledge Graphs	
	Supervisors: prof. Frank van Harmelen (Vrije Universiteit Amsterdam), prof. Paul Groth (University of Amsterdam), Dr. Michael Cochez (Vrije Universiteit Amsterdam).	
	University of Amsterdam	September 2017 - August 2019
	<i>MSc in Artificial Intelligence, cum laude</i>	Amsterdam, The Netherlands
	Thesis: A Modular Framework for Unsupervised Graph Representation Learning	
	Supervisor: Dr. Thomas Kipf	
	UD Francisco José de Caldas	August 2010 - August 2016
	<i>BSc in Electronics Engineering</i>	Bogotá, Colombia
	Thesis: Neural Networks for Learning and Optimizing Electrical Machine Surrogates.	
Current employment	Vrije Universiteit Amsterdam	January 2026 - Present
	<i>Postdoctoral Researcher</i>	Amsterdam, Netherlands
I do research on foundation models for knowledge graphs, simulation-based model validation, and applications to drug-induced kidney disease.		
Work experience	Amsterdam UMC	July 2024 - December 2025
	<i>Postdoctoral Researcher</i>	Amsterdam, Netherlands
	In collaboration with Accenture and clinicians, I carried out research in the field of machine learning for knowledge graphs, and its applications to rare diseases in the healthcare domain.	
	Vrije Universiteit Amsterdam	November 2019 - June 2024
	<i>PhD Researcher</i>	Amsterdam, Netherlands
	I carried out research in representation learning for knowledge graphs, and in relation to the same topic I supervised bachelor and master students, and contributed to course organization and lectures.	
	Bosch Research Center for Artificial Intelligence	May 2023 - August 2023
	<i>Research Intern</i>	Renningen, Germany
	Industrial research internship on the topics of unsupervised learning on graphs for explainable similarity search.	

Irreto B.V.

Data Science Intern

June 2018 - August 2018

Hoofddorp, The Netherlands

Industrial internship on the development and deployment of scalable machine learning models with Tensorflow on Kubernetes clusters and cloud storage, applied to the problem of fraud detection.

Teaching

Machine Learning for Graphs (MSc), Vrije Universiteit Amsterdam January 2026
Contributed two lectures on “Deep Graph Representation Learning” and “Complex Query Answering.”

Deep Learning (MSc), Vrije Universiteit Amsterdam November 2024
Contributed one lecture on “Learning with Graphs.”

Machine Learning for Graphs (MSc), Vrije Universiteit Amsterdam January 2023
Contributed to the design of the first edition of this course, now a recurring MSc course at VU Amsterdam; contributed one lecture on “Approximate Query Answering over Knowledge Graphs”; designed one homework assignment; guided student working sessions; and graded homework.

AI in Practice: Preparing for AI (MOOC), TU Delft
Invited during my PhD to contribute a talk on machine learning for knowledge graphs.

Supervision

Master Students

Paul Constantinescu, “*Zero-shot Entity Disambiguation with Language Models and Search Indexes*”, (MSc thesis, Vrije Universiteit Amsterdam, ongoing).

Akshaye Iyer, “*Neuro-Symbolic Knowledge Graph Construction using Natural Language Understanding*”, (MSc thesis, Vrije Universiteit Amsterdam, ongoing).

Baradwaj Varadharajan, “*Inductive Link Prediction over Novel Relations*” (MSc thesis, University of Amsterdam, 2023).

Qingzhi Hu, “*Data Integration and Predictive Modeling for Impact Investing*” (MSc thesis, University of Amsterdam, 2022).

Stefan Schouten, “*Incorporating Semantics in Knowledge Graph Embeddings*” (MSc thesis, University of Amsterdam, 2021), with Thiviyan Thanapalasingam.

Bachelor Students

Vencel Kuba, “*Evaluating Analogical Reasoning in Knowledge Graph Foundation Models*”, (BSc thesis, Vrije Universiteit Amsterdam, ongoing with Hossein Khojasteh).

Kate Jermakova, “*Structure-Aware Query Corruption in Neural Knowledge Graph Reasoning*”, (BSc thesis, Vrije Universiteit Amsterdam, 2025).

Sawek Mczycki, “*Enhancing Link Prediction in Knowledge Graphs Through Pre-Informed Training*” (BSc thesis, Vrije Universiteit Amsterdam, 2025).

Fredrik Skjelvik, “*Complex Query Answering in the Biomedical Domain*” (BSc thesis, Vrije Universiteit Amsterdam, 2022).

Communication and Outreach	AI & Mathematics Network , Tilburg, The Netherlands	June 2025
	Learning on Knowledge Graphs for Scientific Discovery	
	ELLIS workshop on Representation Learning and Generative Models for Structured Data , Amsterdam, The Netherlands	February 2025
	Interactive Query Answering on Knowledge Graphs with Soft Entity Constraints	
	Austrian Institute of Technology , Vienna, Austria	February 2024
	Learning on Graphs via Multimodal Data	
	Deloitte , Amsterdam, The Netherlands	June 2022
	Learning Entity Representations from Knowledge Graphs and Textual Descriptions	
	Zeta Alpha , Amsterdam, The Netherlands	September 2021
	Inductive Entity Representations from Text via Link Prediction	
Grants and Distinctions	King's College London , London, UK	March 2021
	Complex Query Answering with Neural Link Predictors	
	Elsevier , Amsterdam, The Netherlands	February 2020
	Message Passing Query Embedding	
	Public communication	
	ICT.OPEN video: <i>Discovering Missing Links in Knowledge</i> , public-facing video about research in our lab	2022
	https://youtu.be/mF94YDg4KPE	
	Gold Reviewer	ICML, 2026
	Young Researcher and Innovator Conference Grant	COST Action CA23147, 2026
	Awarded 1500 for participation in the <i>International Conference on Artificial Intelligence in Medicine (AIME 2026)</i> .	
Academic Service	Computing Time on National Computing Facilities (lead co-applicant)	NWO, 2025
	Project: <i>Large Laboratory Models</i> , awarded 300k CPU+2M GPU compute credits.	
	Doctorate cum laude (top 5%)	Vrije Universiteit Amsterdam, 2024
	Best Paper Honorable Mention	Learning on Graphs Conference, 2024
	Outstanding Paper Award	ICLR, 2021
	Reviewing	
	International Conference on Learning Representations (ICLR)	2025, 2026
	Learning on Graphs Conference	2025
	Advances in Neural Information Processing Systems (NeurIPS)	2025, 2026
	International Conference on Machine Learning (ICML)	2025, 2026
Tutorials	The Web Conference	2024, 2025, 2026
	ACM Transactions on Knowledge Discovery and Data	2023
	ACL Workshop on Structured Predictions for NLP	2022
	Conference on Information and Knowledge Management (CIKM)	2022
	Semantic Web Journal	2020, 2025, 2026

Reasoning beyond Triples: Recent Advances in Knowledge Graph Embeddings (CIKM 2023).

Publications

*Indicates equal contribution.

2026

Interactive Query Answering on Knowledge Graphs with Soft Entity Constraints, in **TMLR**.
D. Daza, A. Bernardi, L. Costabello, C. Gueret, M. Mansoury, M. Cochez, M. Schut.

Contextualized Interpretation of Machine-Learning Predictions in Rare Disease Omics: Integrating SHAP, Biomedical Knowledge, and Language Models, in **International Conference on Artificial Intelligence in Medicine**.

D. Daza, Y. Jaspers, A. Bernardi, L. Costabello, C. Gueret, M. Cochez, M. Engelen, S. Kemp, M. Schut.

Explaining Graph Neural Networks for Node Similarity on Graphs, in **TMLR**.

D. Daza, C.X. Chu, T.K. Tran, D. Stepanova, M. Cochez, P. Groth.

Counting Still Counts: Understanding Neural Complex Query Answering Through Query Relaxation, in **TMLR**.

Y. Brunink*, D. Daza*, Y. He, M. Cochez.

Similarity-Constrained Reweighting for Complex Query Answering on Knowledge Graphs, **Under review**.

D. Daza, A. Bernardi, L. Costabello, C. Gueret, M. Cochez, M. Schut.

2025

Discovering Association Rules in High-Dimensional Small Tabular Data, in **ECAI 2025 Workshop on Advanced Neuro-Symbolic Applications**.

E. Karabulut, D. Daza, P. Groth, V. Degeler.

EMERGE: A Benchmark for Updating Knowledge Graphs with Emerging Textual Knowledge, **Under review**.

K. Zaporjets, D. Daza, E. Barba, I. Assent, R. Navigli, P. Groth.

GRAPES: Learning to sample graphs for scalable graph neural networks, in **TMLR**.

T. Younesian, D. Daza, E. van Krieken, T. Thanapalasingam, P. Bloem.

2024

UnRavL: A Neuro-Symbolic Framework for Answering Graph Pattern Queries in Knowledge Graphs, in **Learning on Graphs**.

 Honorable Mention for Best Paper

T. Cucumides, D. Daza, P. Barcelo, M. Cochez, F. Geerts, J.L. Reutter, M.R. Orth.

2023

BioBLP: a modular framework for learning on multimodal biomedical knowledge graphs, in **Journal of Biomedical Semantics**.

D. Daza, D. Alivanistos, P. Mitra, T. Pijnenburg, M. Cochez, P. Groth.

Adapting Neural Link Predictors for Data-Efficient Complex Query Answering, in **NeurIPS**.
E. Arakelyan, P. Minervini, D. Daza, M. Cochez, Isabelle Augenstein.

Harnessing the Web and Knowledge Graphs for Automated Impact Investing Scoring, in **KDD Workshop on AI for Climate Sustainability**.

Q. Hu, D. Daza, L. Swinkels, K. Sait, R. Hoen, and P. Groth .

2022

SlotGAN: Detecting Mentions in Text via Adversarial Distant Learning, in **ACL Workshop on Structured Prediction for NLP**.

D. Daza, M. Cochez, and P. Groth.

2021

Complex Query Answering with Neural Link Predictors, in **ICLR**.

🏆 *Outstanding Paper Award* (top 1%)

E. Arakelyan*, D. Daza*, P. Minervini*, and M. Cochez.

Entity Representations from Text via Link Prediction, in **The Web Conference**.

D. Daza, M. Cochez, and P. Groth.

Approximate knowledge graph query answering: from ranking to binary classification, in **ECAI 2020 Workshop on Graphs for Knowledge Representation and Reasoning**.

R. van Bakel, T. Aleksiev, D. Daza, D. Alivanistos, and M. Cochez.

2020

Message passing query embedding, in **ICML 2020 Workshop on Graph Representation Learning**.

D. Daza and M. Cochez.